



Cairo University
Faculty of Engineering
Credit Hours System
Communications and Computer Engineering Department

Fall 2026

Course Code EECS 304
Course Title Control I
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Sheet 1: Laplace Transform

Question (1):

Solve the following differential equations:

- (a) $\ddot{y}(t) + 3\dot{y}(t) + 2y(t) = u(t)$ = unit step, assume the initial conditions: $y(0) = -1$, $\dot{y}(0) = 2$
(b) $\ddot{y}(t) + 3\dot{y}(t) + 2y(t) = \dot{x}(t) + 3x(t)$, assume the initial conditions: $y(0) = 1$, $\dot{y}(0) = 0$ and the input is given by: $x(t) = e^{-4t}$

Question (2):

Test the linearity of the systems described by the following i/p – o/p relations:

- (a) $y(t) = au(t)$, where ‘a’ is a constant.
(b) $y(t) = u^3(t)$
(c) $y(t) = e^{u(t)}$
(d) $\ddot{y}(t) + a\dot{y}(t) + y(t) = u(t)$, $y(0) = \dot{y}(0) = 0$

Question (3):

Find the Transfer Function of the following systems:

- (a) $\ddot{y}(t) + 3\dot{y}(t) + 2y(t) = \dot{x}(t) + 3x(t)$
(b) $\dot{y}(t) + y(t) = x(t - T)$